causalgeo: concurrent geo holdouts for bundle lift and calibrated channel responses

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Abstract

Marketing teams sometimes turn off different bundles of channels in different groups of markets at the same time. A geo holdout estimates the lift of each bundle. It does not say how to split that lift across the channels inside the bundle. Existing geo tools stop at the bundle. A marketing-mix model can split it, but the split is unstable when the channels move together.

`causalgeo` is a package for concurrent geo holdouts that reports bundle lift, uses impression paths for a model-assisted channel split, and refuses channel estimates that the design cannot distinguish. One bundled test estimates that bundle's total lift. It does not estimate how many sales each channel's impressions were worth. On one constructed series the recovered responses are within 0.21 percent of the known values.

Keywords: geo experiment, incrementality, holdout, marketing mix, Python

1. Motivation and significance

`causalgeo` is a package for concurrent geo holdouts that reports bundle lift, uses impression paths for a model-assisted channel split, and refuses channel estimates that the design cannot distinguish.

A geo test estimates incrementality by comparing markets put on a media change with similar markets left on business as usual [1, 2]. That comparison avoids the bias in lining up people who saw an ad with people who did not [3, 4]. The question is simple: did the channel create incremental sales? [5].

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Table 1: Code metadata.

Nr.	Code metadata description	Value
C1	Current code version	0.2.2
C2	Permanent link to code/repository used for this code version	`https://github.com/adityapt/causalgeo/releases/tag/v0.2.2`
C3	Permanent link to Reproducible Capsule	`https://doi.org/10.5281/zenodo.23787152`
C4	Legal Code License	MIT
C5	Code versioning system used	git
C6	Software code languages, tools, and services used	Python
C7	Compilation requirements, operating environments and dependencies	Python $\geq$ 3.10. Runtime dependencies: pandas, numpy, scipy, statsmodels. The example also uses matplotlib.
C8	Link to developer documentation	`https://github.com/adityapt/causalgeo/blob/main/README.md`
C9	Support email for questions	puttaparthy.aditya@gmail.com

The number of markets that can go dark is usually limited. Take a United States campaign across 210 designated market areas (DMAs). The business may allow a pause in only 30. Large markets may be too risky, some regions may have partner commitments, and others may already be in a flight. Those rules set the holdout pool before anyone looks at balance or power. Advertising lift is small next to ordinary sales movement [6], so one well-powered single-channel test can use the whole pool. Teams therefore test bundles in several cells at once. Each cell estimates the bundle lift, not the lift of the channels inside it.

GeoLift and Meridian GeoX can design and estimate more than one cell [7, 8]. They do not split a cell into channels, and they do not tie a marketing-mix model to the measured cell lift. A mix model estimates channel response from impression history, and channels that move together make that split arbitrary [9, 10]. Robyn and PyMC-Marketing can pull a mix model toward a lift test, including a test of more than one channel [11, 12]. Pulling toward

a test can steady a split [13]. It cannot create information the test and the impression data do not contain. The split below also assumes the bundle lift is the sum of the channel lifts. If the channels interact, these equations do not name the separate responses [14].

The package estimates the bundle lift from the holdout, then uses that lift to guide a straight-line model of channel impressions. The channel responses use both the test and the model. They are not channel-level holdout estimates. If the impressions cannot separate the channels, the package reports the bundle and stops. It does not fill the gap with a sales-per-impression weight of its own. It does not control for in-flight spend, because going dark is what changes that spend [15].

2. Software description

2.1. Software architecture

The workflow behind `design_concurrent_test` and `disentangle` has four parts.

1. `design.py` assigns markets inside groups of similar markets, builds the cell-by-channel flight schedule, and says which channels can be split.
2. `identified.py` estimates one bundle lift per cell, and a channel lift only when that channel was the only one turned off.
3. `exposure.py` checks whether a channel that was supposed to stay on delivered differently in the dark markets and the business-as-usual markets.
4. `calibrated.py` fits the impression response and uses each measured bundle lift to guide the channel split.

The bundle estimate takes out each market's own baseline and each week's common movement, repeating that removal when a market misses some weeks. The uncertainty is by market: weeks inside one market count as one market, not as separate observations [16, 17, 18]. This example has 40 markets and 8 dark markets in each cell. On a separate series of that size, over 30 weeks, redrawing the assignment 200 times, the stated 95 percent intervals covered the true cell lift 91.8 percent of the time. For a single cell that is already assigned, `analyze` can also run time-based regression [19], synthetic control [20, 21], or synthetic difference-in-differences [22]. GeoLift and GeoX still own market selection and the power search [7, 8].

2.2. Software functionalities

Design. The user supplies the markets, groups of similar markets, the channel list, and which channels go dark in each cell. Inside each group, `control_share` stays on business as usual. The other markets are split across cells. `design_from_assignment` takes a cell map that already exists, with the same cell labels. The schedule marks $A_{kc} = 1$ when cell k turns channel c off. Each channel is then called not tested, tied, separate, or not separable. A channel the flight never turns off stays not tested and gets no response.

Bundle lift. $D_{it}^{(k)} = 1$ when market i is in cell k and week t is in the test window. The cell estimate is

$$Y_{it} = \sum_{k} \tau_k \, D_{it}^{(k)} + \alpha_i + \gamma_t + \varepsilon_{it}. \tag{1}$$

τ_k is that cell's incremental sales, after each market's baseline and each week's common movement. It is a channel estimate only when the cell turned one channel off. Overlapping bundles can cover every channel and the bundle lift still belongs to the bundle, not to one channel inside it.

Channels left on. A channel left on in cell k can still deliver differently in the dark markets than in the business-as-usual markets. That impression gap is

$$S_{kc} = \big(\bar{X}_{c,k,\text{post}} - \bar{X}_{c,k,\text{pre}}\big) - \big(\bar{X}_{c,\text{control,post}} - \bar{X}_{c,\text{control,pre}}\big). \tag{2}$$

The pre-test weight b_c is the joint response of the channels, not an estimate of sales on one channel. The published cell lift stays τ_k. If $|b_c S_{kc}|$ is more than `material` times the larger of $|\tau_k|$ and the usual week-to-week sales movement before the test, the channel split uses a refit that accounts for those still-on impressions. `material` is 0.10 unless the user changes it. The user can also publish that refit. Channels that were turned off are the test. They are not controls. The example table shows the requested refit.

Channel split. W_{kc} is the impressions of channel c held out of cell k,

$$W_{kc} = \max(-S_{kc},\, 0) \tag{3}$$

for channels that cell turned off. Every other entry in that row is zero. The mix model should make $\sum_c \beta_c W_{kc}$ close to $-\tau_k$, where β_c is sales per impression. That row is a weighted pull toward the lift test. It is not a hard

constraint. The impression history uses every pre-period week, and only the business-as-usual markets during the flight. Dark market-weeks stay out of that history. A more precise bundle lift pulls harder. The bundle table shows whatever gap is left.

The reported uncertainty treats the cell lifts as separate. It does not fully account for shared control markets or weeks that enter more than one estimate. The simulations check that shortcut on this design, not on every flight.

Impression paths closer than $1 - \texttt{copy_tolerance}$ are treated as copies. The default tolerance is 10^{-6}. One copy stays in the fit so their shared delivery is not ignored, and neither copy gets its own response. By default a response is marked unstable if it moves when the calibration pull is weakened, or when the market with the largest in-flight sales is left out. The tables below turn that check off, so the numbers are the point estimates. A blank impression stays blank, that market-day is dropped, and the fit warns. Daily impact is delivered impressions times $\hat{\beta}_c$. The package can apply geometric adstock. It does not fit a saturation curve, because one on/off holdout cannot trace the response curve [23, 10].

One bundle. One cell measures one number: the bundle lift τ_k. The held-out impressions W_{kc} say how many impressions of each channel were removed. They do not say how many incremental sales each removed impression was worth. Call that unknown weight p_c, in sales per impression. The holdout gives one equation,

$$\sum_c p_c W_{kc} = -\tau_k. \tag{4}$$

Several channels in the same bundle leave this one equation short. The package does not estimate p_c.

If the user already has p_c, from an older single-channel test or from a mix model they are willing to own, the measured lift can be split as

$$a_{kc} = \tau_k \, \frac{p_c W_{kc}}{\sum_j p_j W_{kj}}. \tag{5}$$

The channel pieces add back to the cell lift. Setting every p_c equal to 1 splits the bundle by held-out impressions. Any other weight is an assumption the user brings in. On the public Meridian impression paths, with all five channels dark together and sales built from known weights, an impression

split gives Channel2 7.8 percent of the bundle against a true share of 2.4 percent, and Channel1 12.7 percent against 19.2 percent. Putting those known weights into (5) closes the gaps. The check uses the same weights that built the sales. The holdout does not discover them. The script is `examples/public_allocation_benchmark.py`.

The multi-cell fit above is a different calculation. Different cells turn off different channels, so the impression paths can separate the responses. When they cannot, the response stays blank. Equation (5) is a labeled split of one measured lift. It is not a channel incrementality estimate.

The output is the channel responses, the measured lift next to the lift implied by those responses, the still-on impression check, and daily impact. A market bootstrap reruns the whole pipeline on markets redrawn inside each cell.

3. Illustrative examples

No public geo holdout comes with a known channel-level incrementality. The example uses Google's published Meridian geo file `geo_all_channels.csv`: 40 markets and 156 weeks, from 25 January 2021 through 15 January 2024 [24]. The public file demonstrates that the software runs. The sales are constructed so the channel responses are known. That is a check of the software, not evidence that a campaign worked. Meridian's conversions column is used only to group similar markets. It is not the outcome, because that series already contains Meridian's own media response.

Three of the five channels are used, with planted responses 2×10^{-4}, 5×10^{-4}, and 1×10^{-4} sales per impression. Markets are grouped into quartiles of average conversions. With `control_share = 0.4` and seed 1, 16 markets stay on business as usual and 8 go into each cell. The flight is the last 16 weeks, starting 2 October 2023. In a dark market-week the turned-off channel's impressions are set to zero, and sales are

$$Y_{it} = \alpha_i + \gamma_t + \sum_c \beta_c X_{itc} + \varepsilon_{it}, \tag{6}$$

with $\alpha_i \sim N(0, 50^2)$, $\gamma_t \sim N(0, 20^2)$, and $\varepsilon_{it} \sim N(0, 30^2)$, drawn with seed 7. The script is `paper/example.py`.

The cells go dark on {Channel0, Channel1}, {Channel0, Channel2}, and {Channel1, Channel2}. No channel is dark alone, so the holdout-only table

Table 2: Cell lift on the constructed series, with the still-on refit requested. By default the unadjusted holdout stays the published number and the impression shift is only recorded. The model column values the dark-channel impression change at the fitted responses.

Cell	Bundle	Unadjusted	Adjusted	SE	Model
1	Channel0+Channel1	−555.7	−555.7	62.9	−570.2
2	Channel0+Channel2	−318.0	−251.5	36.2	−253.6
3	Channel1+Channel2	−399.9	−399.9	54.4	−396.2

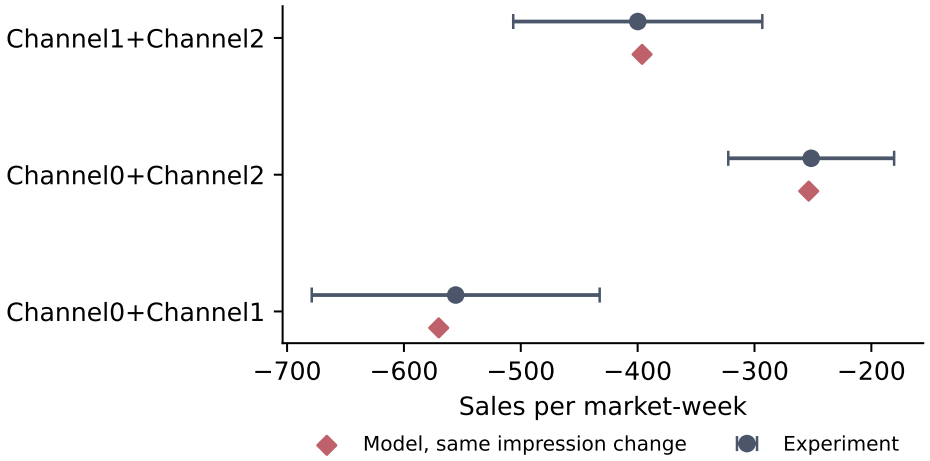


Figure 1: Measured bundle lift, with a 95 percent interval, and the same dark-channel impression change valued at the fitted responses. Sales per market-week.

leaves all three channel estimates blank. Table 3 reports the impression-model responses, pulled toward the three measured bundle lifts.

This checks whether the package recovers the straight-line response it was built to fit. It is not evidence about saturation, long carryover, or whether eight markets are enough to power a cell. A real flight with a smaller lift will have a wider interval [6].

Cell 2 shows the still-on check. Channel1 was supposed to stay on, but its impression gap, valued at the joint pre-period response, is worth −59 sales, or 19 percent of the unadjusted lift. Refitting moves the estimate from −318.0 to −251.5. Cells 1 and 3 stay on the unadjusted holdout.

Table 2 puts the holdout lift next to the lift implied by the channel responses. The gaps are 2.6, 0.8, and 0.9 percent of the holdout lifts, and

Table 3: Planted and recovered response, in sales per 10,000 impressions. The largest absolute error is 0.21 percent.

Channel	Planted	Recovered	SE	Abs. error (%)
Channel0	2.000	1.997	0.006	0.17
Channel1	5.000	5.009	0.008	0.17
Channel2	1.000	1.002	0.008	0.21

Table 4: Two hundred refits per setting. Rho is how far the live Channel1 impression path is mixed toward Channel0, and the correlation is what that mix produces. Noise multiplies the example's leftover sales movement. Error is the median absolute percent error among reported responses. The brackets are the range of that coverage across the simulated tests. At rho 1, error and coverage refer to the channel that is still separate.

Noise	Rho	Corr.	Runs	Error (%)	Coverage [MC interval]	Withheld
1	0.0	0.34	200	0.19	0.958 [0.942, 0.974]	0.00
1	0.9	0.96	200	0.40	0.953 [0.934, 0.973]	0.00
1	1.0	1.00	200	0.58	0.940 [0.907, 0.973]	1.00
4	0.0	0.34	200	0.76	0.958 [0.942, 0.974]	0.00
4	0.9	0.96	200	1.62	0.957 [0.938, 0.975]	0.00
4	1.0	1.00	200	2.25	0.940 [0.907, 0.973]	1.00

every model total sits inside the holdout interval. The bundle number is the holdout. The channel number is the model, pulled toward that holdout.

The same three cell lifts and the held-out impressions can be solved for the three channel responses with no pre-period impression history. That holdout-only split gives 2.22, 4.48, and 2.01 sales per 10,000 impressions, with uncertainty 0.43, 0.64, and 1.14. The calibrated uncertainty on the same series is 0.006, 0.008, and 0.008. The tighter intervals come from the impression history, not from the three cell lifts.

When Channel1 is replaced with a copy of Channel0, both stay blank. Their shared path remains as one control, and Channel2 comes back as 0.97 sales per 10,000 impressions against a planted 1.00. Dropping that shared path instead pushes it to 3.02. Figure 2 shows those two fits against the planted response.

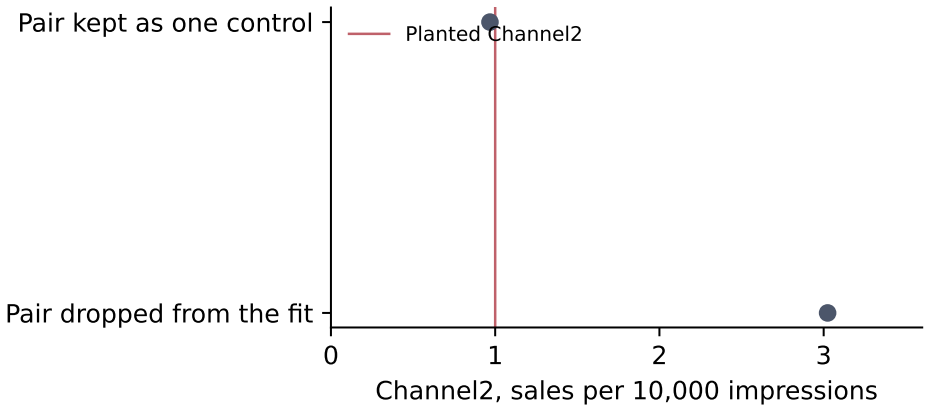


Figure 2: Channel2 when Channel1 is an exact copy of Channel0. Keeping the shared delivery as one control recovers the planted response. Dropping that path pushes the remaining response from about 1 to about 3 sales per 10,000 impressions.

Table 4 and Figure 3 summarize 200 draws for each mix and noise level. At the example noise, median absolute error is 0.19 percent when the impression paths are distinct, where their correlation is 0.34, and 0.40 percent when the mix weight is 0.9 and the correlation is 0.96. The 95 percent intervals cover the truth 95.8 and 95.3 percent of the time. At an exact copy, the pair is left blank in every run and the remaining channel is covered in 94.0 percent of runs. Multiplying the leftover sales noise by four raises the distinct-path error from 0.19 to 0.76 percent and leaves coverage at 95.8 percent. These checks are for this design, and for uncertainty that treats the cells as separate.

Table 5: Stress test over 200 draws when the sales response is not the straight line the package fits. Error is the median absolute percent error in each channel's contribution to the holdout contrast. Coverage is how often the stated 95 percent interval contains that contribution. The brackets are the range of that coverage across the draws. Gap is the median absolute difference between the model bundle and the holdout lift, as a percent of the holdout lift.

Fit	Runs	Error (%)	Coverage [MC interval]	Gap (%)
Carryover omitted	200	47.27	0.000 [0.000, 0.000]	48.09
Carryover modeled	200	0.19	0.942 [0.922, 0.961]	2.22
Saturation omitted	200	26.99	0.000 [0.000, 0.000]	26.97

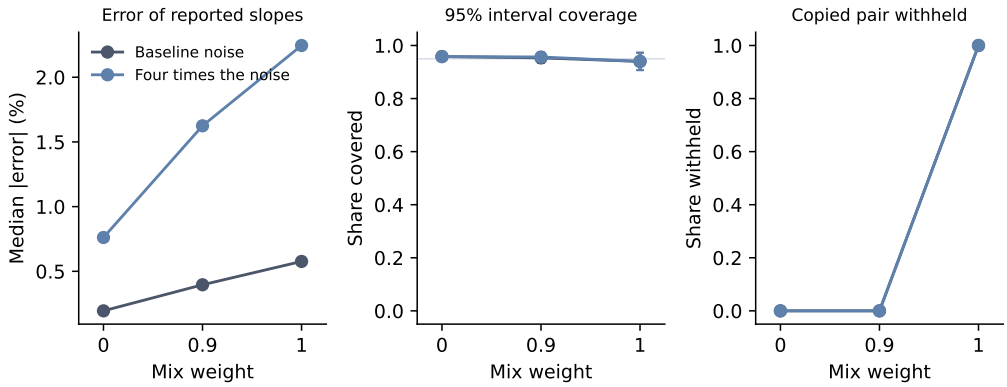


Figure 3: Recovery over 200 simulations as the live Channel1 impression path is mixed toward Channel0. The second series multiplies leftover sales noise by four. The bars are the range of coverage across those simulations. At mix weight 1 the pair is left blank, so error and coverage refer to the channel that is still separate.

The recovery check above uses the same straight-line response that built the sales. Table 5 instead turns on two responses the fit can miss, over 200 draws. Carryover keeps half of last week's impressions in this week's effect. Saturation uses $\beta_c X/(1+X/h_c)$, where h_c is twice the median positive delivery and β_c is the response at the first impression. Neither response has one true sales-per-raw-impression number, so the error is judged against each channel's known contribution to the same dark-versus-BAU contrast as the holdout. The carryover case is also refit with the decay that built the sales, to check the optional adstock.

Putting that decay in brings the error back to the straight-line result. Leaving carryover or saturation out produces large channel errors, and the 95 percent intervals miss the truth every time. The bundle gap is large too, so the package shows the disagreement instead of making a bad channel split look like a clean lift estimate. These checks do not pick a decay, and they do not fit a saturation curve.

Table 6: Further checks, 30 draws each, so the coverage ranges are wide. Error is the median absolute percent error among reported responses. The confounded row reports the Channel0 error. The brackets are the range of coverage across the draws. Serial correlation means this week's leftover sales still look like last week's, with correlation 0.8. Market-specific response lets each market sit between half and one and a half times the planted response. Redistribution moves half of a typical held-out week onto a channel the cell left on. Incomplete delivery leaves 30 percent of the business-as-usual level still running. The confounder follows Channel0's usual delivery and does not go dark. The collinear rows mix Channel1 toward Channel0 with weight 0.999, correlation 0.9995, and responses at 5 percent of the example.

Setting	Runs	Error (%)	Coverage [MC interval]	Gap (%)
Serial correlation	30	0.43	0.956 [0.914, 0.997]	3.56
Heterogeneous slopes	30	4.81	0.344 [0.248, 0.441]	4.01
Budget redistributed	30	0.22	0.967 [0.930, 1.000]	12.60
Redistribution, refit requested	30	0.22	0.967 [0.930, 1.000]	12.60
Incomplete delivery	30	0.22	0.967 [0.930, 1.000]	2.83
Observational fit only	30	0.22	0.967 [0.930, 1.000]	2.12
Calibrated fit	30	0.22	0.967 [0.930, 1.000]	2.13
Confounded Channel0	30	49.94	0.644 [0.614, 0.675]	21.12
Collinear, observational	30	44.64	0.956 [0.895, 1.000]	43.33
Collinear, calibrated	30	10.52	0.989 [0.967, 1.000]	8.79
Experiment only	30	11.16	1.000 [1.000, 1.000]	–

Table 6 asks what that recovery depends on. Sales that linger from week to week, and a recorded shortfall in delivery, leave the channel error near the straight-line result. A response that differs by market does not: coverage falls to 34.4 percent. Moving the lost impressions onto a channel left on leaves the channel error near 0.2 percent. The raw cell lift sits 52.9 percent away from the lift after those extra impressions are removed, because the holdout still contains the moved delivery. The channel split is calibrated to the cleaned lift, and the remaining bundle gap is 12.6 percent. Publishing that refit instead of the raw cell lift does not change the split, so the two budget rows agree. On the original series the impression-history fit and the calibrated fit both have 0.22 percent error. The calibration rows are not what produces that recovery.

A confounder that tracks Channel0 and stays on during the holdout leaves that response 49.9 percent high after calibration, against 50.0 percent from the impression history alone. The bundle gap opens to 21 percent. The gap moves. The response does not. Calibration does change the split when the channels are nearly copies and the planted responses are small. At a mix

weight of 0.999 the impression-history error is 44.6 percent and the calibrated error is 10.5 percent, and the bundle gap falls from 43 percent to 8.8 percent. Both coverage ranges come from only 30 draws and are wide: the impression-history range runs from 0.895 to 1, and the calibrated range runs from 0.967 to 1. The holdout-only split, on the original straight-line series, has median error 11.2 percent and coverage 1 because its intervals are wide. Figure 4 puts coverage and the bundle gap next to the carryover and saturation checks.

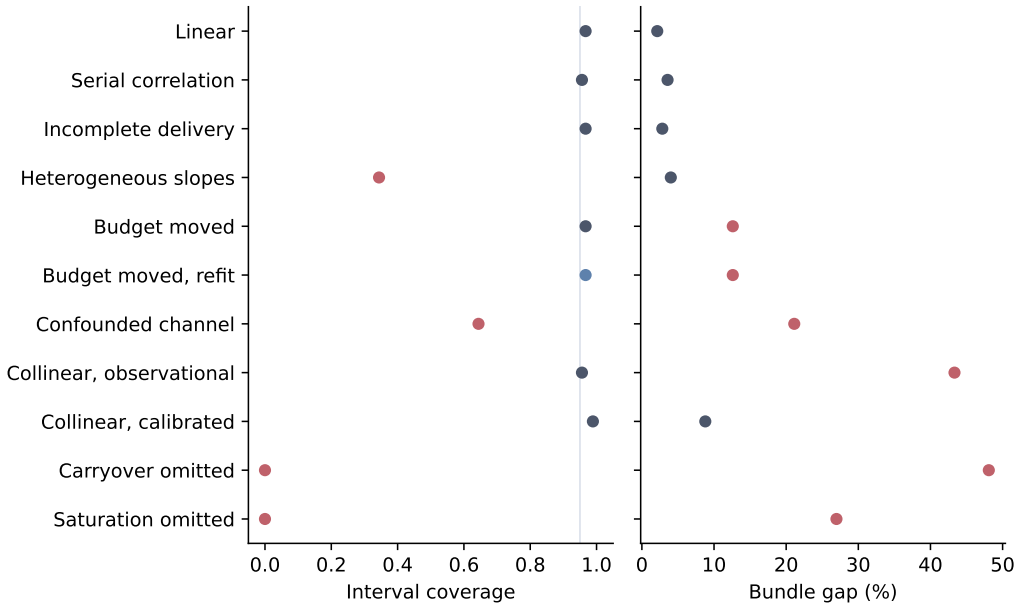


Figure 4: Coverage and the bundle gap. The straight-line, serial, delivery, market-response, budget, confounder, and collinear rows are 30 draws. Carryover and saturation are the 200-draw checks in Table 5. The pale line is 95 percent coverage.

4. Impact

`causalgeo` is a package for concurrent geo holdouts that reports bundle lift, uses impression paths for a model-assisted channel split, and refuses channel estimates that the design cannot distinguish. GeoLift and GeoX can run more than one cell. They do not make this split [7, 8]. Robyn and PyMC-Marketing already calibrate a mix model to lift tests [11, 12]. This package is the concurrent holdout, the reported gap, and the blank channel when the impressions cannot separate it. The bundle lift is the holdout. A

channel response is reported when the cells separate the channels. Splitting one bundle with a sales-per-impression weight is an assumption the user brings, and the package does not estimate that weight.

The current release is 0.2.2. Adoption is early, and this paper has no user study. `paper/reproduce.py` rebuilds all five tables and four figures. The same functions apply to a real holdout once the user supplies market outcomes and channel impressions. Channels the flight cannot separate, or whose impressions are copies, stay blank rather than receiving a made-up response.

5. Conclusions

`causalgeo` estimates concurrent channel-bundle holdouts and splits a measured bundle only when the impression paths can separate the channels. One bundled test estimates that bundle's lift. The impressions removed from each channel are observed. The sales per impression are not. Equation (4) is one equation for every channel in the bundle, so those channel estimates stay blank unless other cells separate them. Equation (5) splits the measured lift only after the user supplies the sales-per-impression weights. Those weights are an input. They are not an estimate from the holdout. The calibration is a pull, not a force: it moves the mix model toward the lift test and still shows the gap. On one constructed series the responses are within 0.21 percent of the planted values. Solving the same cell lifts with no impression history gives uncertainty of 0.43 to 1.14 sales per 10,000 impressions, against 0.006 to 0.008 for the calibrated split. Across 200 draws, four times the leftover sales noise raises median error from 0.19 to 0.76 percent and leaves coverage at 95.8 percent. A response that differs by market cuts coverage to 34.4 percent. A confounder the holdout leaves on keeps the affected response about 50 percent high. Nearly copied channels with small effects move from 44.6 percent error to 10.5 percent once the cell lifts are included. A carryover fit that uses the decay from the sales gives 94.2 percent coverage. Leaving out carryover or saturation produces large channel errors and large bundle gaps. On a 30-week series with the same market counts, the cell intervals covered the true lift in 91.8 percent of 200 redrawn assignments. The intervals still depend on the response that was fitted, and on treating the cells as separate.

The split is a straight line, with optional geometric adstock and no saturation curve [10]. Turning a channel on or off does not trace the response curve, so the package does not fit one, and it does not estimate an interac-

tion between channels [14]. GeoLift and GeoX handle market selection and power [7, 8]. This package assigns markets inside groups the user has already chosen, or accepts that assignment, and does not claim that eight markets make a powered cell [6]. The published fits leave the stability check off. The default cell number is the unadjusted holdout. When lost impressions move onto a channel left on, the split uses the refitted cell lift after that delivery is removed. In-flight spend stays out of the estimate because the assignment is what changes it [15, 3].

CRediT authorship contribution statement

Aditya Puttaparthi Tirumala: Conceptualization, Methodology, Software, Validation, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The impression paths are Google's public Meridian file `geo_all_channels.csv` [24]. `examples/meridian_conversions.py` runs the same split on Meridian's conversions column. That run has no known channel incrementality. It only shows the software on that file. The archived v0.2.2 release is `https://doi.org/10.5281/zenodo.23787152`. In `paper/`, set `PYTHONPATH` to `../src` and run `python3.11 reproduce.py`. Set `MPLCONFIGDIR` to a writable directory first if Matplotlib warns about its font cache. The script does that itself when the variable is unset. That rebuilds Tables 2–6 and the four figures, then checks them against the rounded values reported here. `python3.11 reproduce.py --fast` uses two draws per setting and only checks that the scripts finish. Table 1 gives the repository links.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Cursor, with the Grok language model, to draft and revise the manuscript and the example

code. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

References

[1] J. Vaver, J. Koehler, Measuring ad effectiveness using geo experiments, Tech. rep., Google Inc. (2011).
URL https://research.google/pubs/measuring-ad-effectiveness-using-geo-experiments/

[2] J. Vaver, J. Koehler, Periodic measurement of advertising effectiveness using multiple-test-period geo experiments, Tech. rep., Google Inc. (2012).
URL https://research.google/pubs/periodic-measurement-of-advertising-effectiveness-using-multiple-test-period-geo-experiments/

[3] R. A. Lewis, J. M. Rao, D. H. Reiley, Here, there, and everywhere: Correlated online behaviors can lead to overestimates of the effects of advertising, in: Proceedings of the 20th International Conference on World Wide Web, 2011, pp. 157–166. doi:10.1145/1963405.1963431.

[4] B. R. Gordon, F. Zettelmeyer, N. Bhargava, D. Chapsky, A comparison of approaches to advertising measurement: Evidence from big field experiments at Facebook, Marketing Science 38 (2) (2019) 193–225. doi:10.1287/mksc.2018.1135.

[5] T. Blake, C. Nosko, S. Tadelis, Consumer heterogeneity and paid search effectiveness: A large-scale field experiment, Econometrica 83 (1) (2015) 155–174. doi:10.3982/ECTA12423.

[6] R. A. Lewis, J. M. Rao, The unfavorable economics of measuring the returns to advertising, The Quarterly Journal of Economics 130 (4) (2015) 1941–1973. doi:10.1093/qje/qjv023.

[7] Meta Open Source, GeoLift: geo-level incrementality using synthetic control methods (2021).
URL https://github.com/facebookincubator/GeoLift

[8] S. Ye, F. Sinn, Y. Li, Y. Jiao, Meridian GeoX: An open-source framework for precision and efficiency in geo experiments, Google Research (2026).
URL https://research.google/pubs/meridian-geox-an-open-source-framework-for-precision-and-efficiency-in-geo-experiments/

[9] D. Chan, M. Perry, Challenges and opportunities in media mix modeling, Tech. rep., Google Inc. (2017).
URL https://research.google/pubs/challenges-and-opportunities-in-media-mix-modeling/

[10] Y. Jin, Y. Wang, Y. Sun, D. Chan, J. Koehler, Bayesian methods for media mix modeling with carryover and shape effects, Tech. rep., Google Inc. (2017).
URL https://research.google/pubs/bayesian-methods-for-media-mix-modeling-with-carryover-and-shape-effects/

[11] J. Runge, I. Skokan, G. Zhou, K. Pauwels, Packaging up media mix modeling: An introduction to Robyn's open-source approach, arXiv preprint arXiv:2403.14674 (2024).
URL https://arxiv.org/abs/2403.14674

[12] PyMC Labs, PyMC-marketing: Bayesian marketing mix models (2024).
URL https://www.pymc-marketing.io/

[13] Y. Zhang, M. Wurm, E. Li, A. Wakim, J. Kelly, B. Price, Y. Liu, Media mix model calibration with Bayesian priors, Tech. rep., Google LLC (2023).
URL https://storage.googleapis.com/gweb-research2023-media/pubtools/pdf/a09f404fdc3107fafb7a52cc5af6a80e4d0fda2b.pdf

[14] P. A. Naik, K. Raman, Understanding the impact of synergy in multimedia communications, Journal of Marketing Research 40 (4) (2003) 375–388. doi:10.1509/jmkr.40.4.375.19385.

[15] J. D. Angrist, J.-S. Pischke, Mostly Harmless Econometrics: An Empiricist's Companion, Princeton University Press, Princeton, NJ, 2009.

[16] J. M. Wooldridge, Econometric Analysis of Cross Section and Panel Data, 2nd Edition, MIT Press, Cambridge, MA, 2010.

[17] M. Bertrand, E. Duflo, S. Mullainathan, How much should we trust differences-in-differences estimates?, The Quarterly Journal of Economics 119 (1) (2004) 249–275. doi:10.1162/003355304772839588.

[18] A. C. Cameron, D. L. Miller, A practitioner's guide to cluster-robust inference, Journal of Human Resources 50 (2) (2015) 317–372. doi:10.3368/jhr.50.2.317.

[19] J. Kerman, P. Wang, J. Vaver, Estimating ad effectiveness using geo experiments in a time-based regression framework, Tech. rep., Google Inc. (2017).
URL https://research.google/pubs/estimating-ad-effectiveness-using-geo-experiments-in-a-time-based-regression-framework/

[20] A. Abadie, A. Diamond, J. Hainmueller, Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program, Journal of the American Statistical Association 105 (490) (2010) 493–505. doi:10.1198/jasa.2009.ap08746.

[21] Y. Xu, Generalized synthetic control method: Causal inference with interactive fixed effects models, Political Analysis 25 (1) (2017) 57–76. doi:10.1017/pan.2016.2.

[22] D. Arkhangelsky, S. Athey, D. A. Hirshberg, G. W. Imbens, S. Wager, Synthetic difference-in-differences, American Economic Review 111 (12) (2021) 4088–4118. doi:10.1257/aer.20190159.

[23] D. M. Hanssens, L. J. Parsons, R. L. Schultz, Market Response Models: Econometric and Time Series Analysis, 2nd Edition, Kluwer Academic Publishers, Boston, 2001.

[24] Google, Meridian simulated geo media data, file geo_all_channels.csv (2024).
URL https://github.com/google/meridian